

What is Hypothesis Testing?

Definition :

A *hypothesis test* (or test of significance) is a standard procedure for testing a claim or statement about a property of a population.

Null Hypothesis vs Alternative Hypothesis

Null Hypothesis:

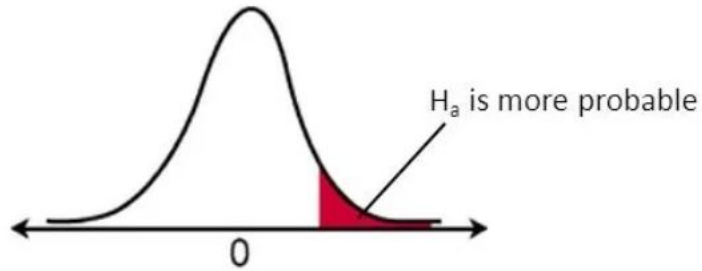
Statement or claim being made which we are trying to disprove.

Alternative Hypothesis:

Is the hypothesis that we are trying to prove and which is accepted if we have sufficient evidence to reject the null hypothesis.

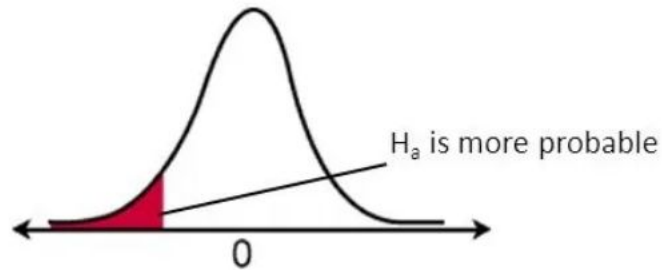
The alternative hypothesis (H_a) is an answer to the research question that suggests that there is a real effect in the study population.

Null Hypothesis vs Alternative Hypothesis



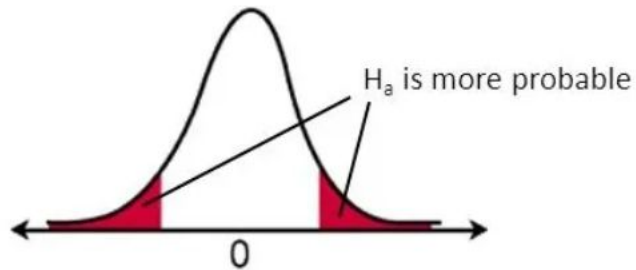
Right-tail test

$$H_a: \mu > \text{value}$$



Left-tail test

$$H_a: \mu < \text{value}$$



Two-tail test

$$H_a: \mu \neq \text{value}$$

Null Hypothesis v Alternative Hypothesis

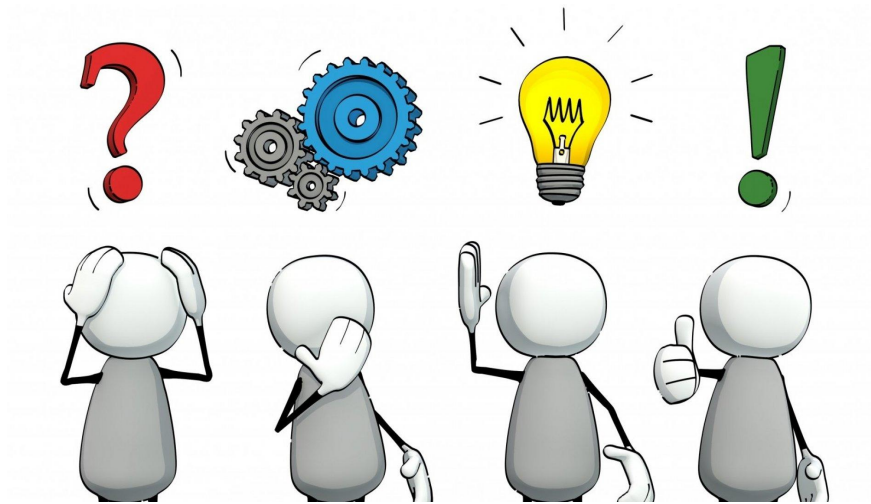
Key Ideas

- ❑ Null Hypothesis is the “no effect” claim while the Alternative is the “effect is present” claim
- ❑ Your research question will help you identify whether you are creating a left tailed, right tailed, or two tailed alternative
- ❑ Let’s see an example of a Null and Alternative Hypothesis for a research question!

Example

- About 10% of the human population is left-handed. Suppose a researcher at Penn State speculates that students in the College of Arts and Architecture are more likely to be left-handed than people found in the general population. We only have one sample since we will be comparing a population proportion based on a sample value to a known population value.
- **Research Question:** *Are artists more likely to be left-handed than people found in the general population?*

What is the Null and Alternative Hypothesis for this example?



Example

- **Research Question:** *Are artists more likely to be left-handed than people found in the general population?*

ANSWER

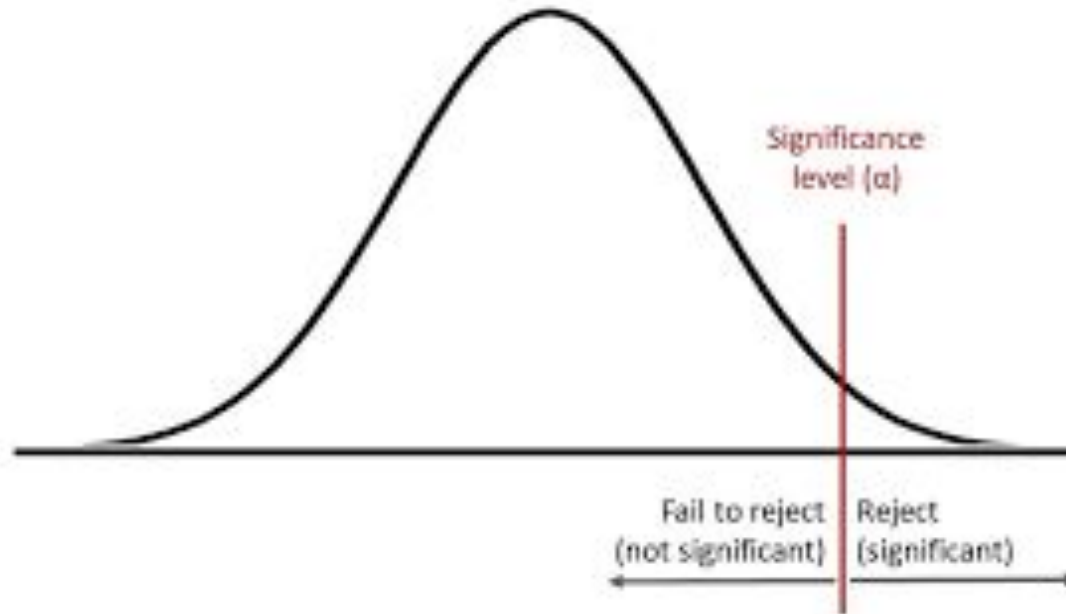
- **Null Hypothesis:** Students in the College of Arts and Architecture are no more likely to be left-handed than people in the general population (population percent of left-handed students in the College of Art and Architecture = 10% or $p = .10$).
- **Alternative Hypothesis:** Students in the College of Arts and Architecture are more likely to be left-handed than people in the general population (population percent of left-handed students in the College of Arts and Architecture $> 10\%$ or $p > .10$). This is a one-sided alternative hypothesis.

What is Significance Level in Hypothesis Testing?

To decide if we have *sufficient* evidence against the null hypothesis to reject it (in favor of the alternative hypothesis), we must first decide upon a *significance level*.

The significance level is the probability of rejecting the null hypothesis when it the null hypothesis is true and is denoted by α . The 5% significance level is a common choice for statistical test.

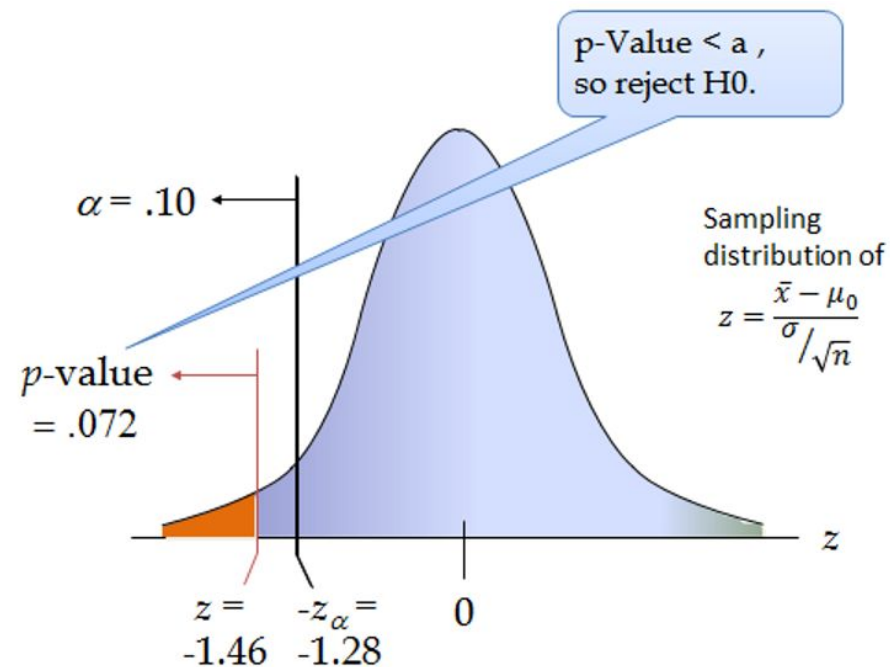
Visualize: Significance Level



In a right tailed test, we would reject the null hypothesis for any values to the RIGHT of the significance level

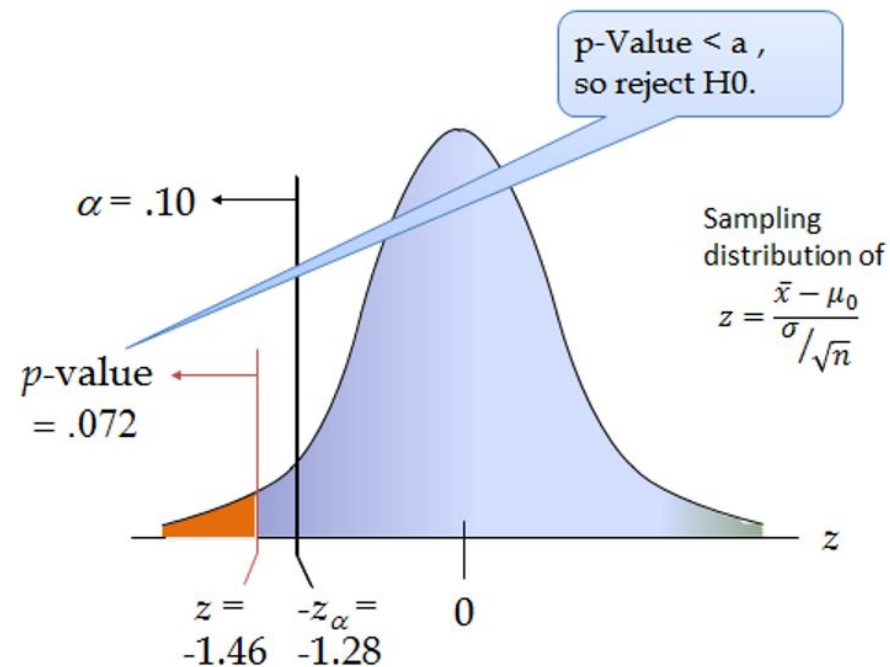
P-Value

The P value is defined as the probability under the assumption of no effect or no difference (null hypothesis), of obtaining a result equal to or more extreme than what was actually observed.



P-Value

Here, we see that a P-value that is SMALLER than our significance level of .1 means that we reject the null hypothesis. This is a LEFT Tailed test



Testing Methods

One Sample Z-Test

- Statistic: Z Statistic
- Dealing with 1 Population and a sample from that population
- Assumptions
 - 1. Mean and variance of the population are known.
 - 2. The test statistic follows normal distribution.

Example

This [Video](#) gives an in-depth look at an example for One Sample Z-tests

ONE SAMPLE Z-TEST

Hypothesis:

$H_0: \mu \leq 7.5 \text{ lbs}$

$H_a: \mu > 7.5 \text{ lbs}$

Number	Mean Weight	σ
35	8.2 lbs	2.1 lbs

Confidence Level = 95%

Alpha level = 5% or .05

1. Calculate Z:
$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$
$$Z = \frac{8.2 - 7.5}{2.1 / \sqrt{35}} = \frac{0.7}{0.355} = 1.97$$
2. Look up P-Value in Z-Table
P-Value = $1 - 0.9756 = 0.0244$
3. Conclusion:
Since P-value < Alpha level, there is sufficient evidence to reject H_0 .

Key Ideas: One Sample Z-Test

- The hypothesis for One Sample Z-Test will look like one of these 3 pairs:

$$H_0 : \mu_1 = \mu$$

$$H_A : \mu_1 \neq \mu$$

$$H_0 : \mu_1 \leq \mu$$

$$H_A : \mu_1 > \mu$$

$$H_0 : \mu_1 \geq \mu$$

$$H_A : \mu_1 < \mu$$

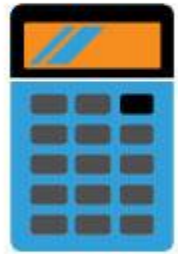
Key Ideas: One Sample Z-Test

- For two-tailed tests, you must look at the Z-Statistic for both the positive and negative alpha values. A portion of the Z score table is below:

<i>z</i>	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9933	.9934	.9936

Key Ideas: One Sample Z-Test

Z Test Statistics Formula



$$\text{Z Test} = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$



Your Turn!

- : A coaching institute claims that the students' mean scores in their institute are greater than the 82 marks with a standard deviation of 20. A sample of 81 students is selected, and the mean score is 90 marks. At 95% confidence level, is there enough evidence to support the claim?

Your Turn!

- State the null and alternative hypothesis
 - Null hypothesis $H_0: \mu \leq 82$
 - Alternative hypothesis $H_1: \mu > 82$
- Select appropriate statistic- Since the claim is student marks are greater than 82 marks; it is a right-tailed test.
- Level of significance: $\alpha = 0.05$
- Find the critical value Z_{critical} : $p\text{-value} = 1 - \alpha = 1 - 0.05 = 0.95$
- Look at the 0.95 in z table $\Rightarrow Z_{\text{critical}} = 1.645$

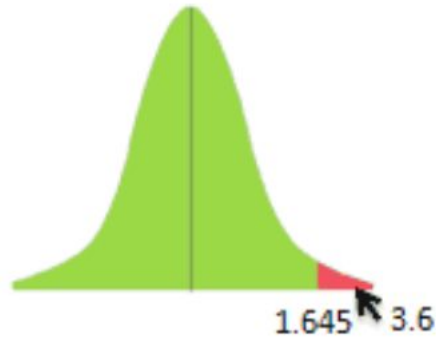
Your Turn!

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2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964

Your Turn!

- Calculate the test statistics

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{90 - 82}{20/\sqrt{81}} = \frac{8}{2.2} = 3.6$$



Your Turn!

- Interpret your results!

Interpret the results: Compare Z_{calc} to Z_{critical} . In hypothesis testing, a critical value is a point on the test distribution compares to the test statistic to determine whether to reject the null hypothesis. Since z_{cal} value is greater than z critical value and it is in the rejection region. Hence we can reject the null hypothesis.

There is enough evidence to support the students' scores in their institute is greater than the 82 marks.